Beyond Acceptance: Disentangling Data Support and Reward Optimization for Natural Language Inference Adaptation

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Abstract

Filtering creates a continuation problem: learning from accepted and rejected populations once a separately frozen instruction-model verifier becomes available. We study this problem through a controlled decomposition of training support, learning objective, and output regularization. Starting from a common filtered NLI model, we compare soft-label cross-entropy, sampled REINFORCE leave-one-out, and exact expected-reward optimization using a frozen six-view instruction-model verifier, without target labels in encoder adaptation. Across three optimization seeds, exact reward improves target macro-F1 by 5.69 points over soft-label cross-entropy, whereas cross-entropy produces better log loss. At a fixed unique-example and update budget, replacing 40% of accepted examples with rejected examples reduces target macro-F1 by 3.06 points while leaving mean cross-task probe performance nearly unchanged. Gradient analysis explains why equal reward information and equal initial gradient magnitude need not produce equal learning dynamics. Paired frozen probes further separate prediction adaptation from transferable representation change. Together, these results establish an evaluation framework for verifier-guided adaptation in which support allocation, objective geometry, and the intended prediction endpoint are explicit design choices.

1 Introduction

Confidence filtering selects unlabeled training inputs, yet discarded examples remain part of the deployment distribution. A frozen instruction-model verifier can revisit them through soft targets, sampled prediction rewards, or exact expected reward. These objectives can produce different outcomes from identical verifier information.

Continuation can change rejected-input exposure, coverage, repetition, and objectives simultaneously, obscuring attribution. After noisy source supervision, decision quality may also differ from information accessible in representations.

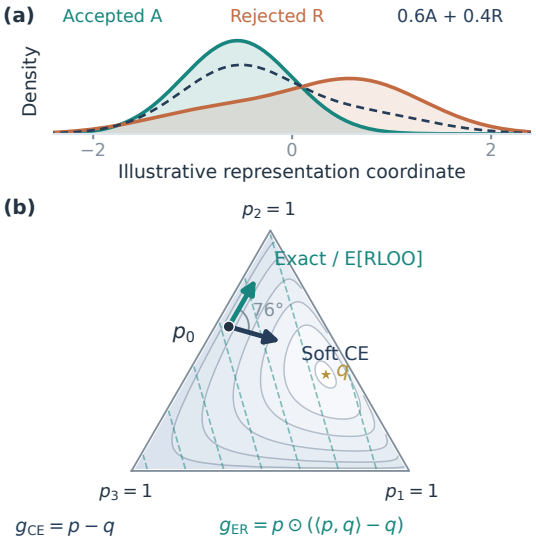


Figure 1: (a) Schematic normalized accepted/rejected densities and their 60/40 replacement mixture. (b) Analytic three-class simplex: navy CE contours and teal reward level sets, with local logit-descent directions mapped through the softmax Jacobian at $p_0 = (0.05, 0.60, 0.35)$ with illustrative verifier distribution $q = (0.50, 0.40, 0.10)$ and $w = 1$. Arrows share one scale; expected RLOO equals Exact. Reference KL has zero gradient when $p_{\text{ref}} = p_0$; consistency is omitted.

We study verifier-guided natural language inference (NLI) adaptation (Figure 1). A Qwen classifier transfers French source supervision to unlabeled English targets through semantic–confidence filtering. A frozen six-view verifier scores three continuation objectives with shared output regularization. Support and initial gradient-magnitude controls yield eight conditions across three seeds; native predictions and frozen probes separate decisions from accessible representation information.

The results identify three distinctions: reward optimization favors decisions while soft CE favors log loss; pool expansion and fixed-budget replacement have different costs; and native-head gains exceed fixed-support probe changes. These contrasts con-

nect objective geometry and exposure allocation to measured adaptation endpoints, without equating verifier agreement and performance.

We contribute a fixed-feedback continuation formulation; propositions separating objective gradients, sampling variance, and regularized optima; and three-seed evidence connecting decisions, probability quality, optimization participation, and transfer under fixed probe supports. Code and protocols are available in the anonymous repository.

2 Related Work

Selection and self-training. Self-training complements pretrained NLU through retrieved unlabeled examples (Du et al., 2021). Confidence thresholds (Sohn et al., 2020) and continuous weights (Chen et al., 2023) trade pseudo-label quality against coverage; confirmation bias motivates controlling their influence (Arazo et al., 2020). We condition on an existing gate and distinguish preserving, expanding, and replacing its support. Fixed-budget replacement isolates the allocation of exposures to accepted and rejected inputs from simply adding more unique examples.

Model feedback and verifier design. AI feedback can provide rewards directly from an instruction model (Lee et al., 2024), while self-rewarding models update their own judging and generation capabilities (Yuan et al., 2024). Confidence-coupled feedback can amplify reward bias (Tan et al., 2026). We freeze and cache a separate verifier, holding feedback evolution fixed. Prompt and answer-order sensitivity motivate explicit view controls (Zhao et al., 2021; Zheng et al., 2023); our agreement weight measures view stability, not verified correctness. The reward concerns three NLI labels, not open-ended response preferences.

Distillation and reward optimization. Distillation fits a teacher distribution by cross-entropy (Hinton et al., 2015); REINFORCE optimizes expected reward with score-function gradients (Williams, 1992). RLOO supplies an action-independent baseline without a learned critic (Ahmadian et al., 2024). Exact enumeration shares its objective but removes action-sampling variance. DPO connects KL-regularized rewards to optimal policies and learns from response preferences (Rafailov et al., 2023); our reward is fixed and fully enumerated, allowing a soft-CE comparison using identical verifier information. This separates estimator variance from objective choice.

Probability quality and representation. Recent RLVR studies report contrasting effects on reasoning coverage (Yue et al., 2025; Liu et al., 2025a). We examine distinct classification endpoints: proper scores (Gneiting and Raftery, 2007), F1, and frozen transfer. Dropout consistency regularizes prediction variation (Liang et al., 2021); we hold its JS form fixed. Probe results depend on decoder capacity and training support (Hewitt and Liang, 2019; Belinkov, 2022). Matched probes separate the deployed classifier from information accessible to a fresh decoder.

Controlled continuation separates objective geometry from exposure allocation under fixed verifier feedback. We derive gradient, estimator, and pointwise-optimum consequences, then connect them to empirical decisions, probability quality, and fixed-support readouts. This formalization holds feedback evolution outside the intervention.

3 Verifier-Guided Continuation

3.1 Fixed target support and initialization

Definition 1 (Fixed-feedback instance)**.** *An instance fixes an unlabeled pool $\mathcal{T} = \{x_i\}_{i=1}^{N}$, a gate $g_i \in \{0, 1\}$, an initial policy π_0, and cached feedback $q_i = q(x_i) \in \Delta^{K-1}$, $w_i = w(x_i) \in [0, 1]$. Here K is the class count and $\Delta^{K-1} = \{p \in \mathbb{R}_+^K : \sum_y p_y = 1\}$. The gate defines $\mathcal{A} = \{i : g_i = 1\}$ and $\mathcal{R} = \{i : g_i = 0\}$, shared by continuation methods within each student seed.*

The policy π_0 is adapted from source supervision using accepted pseudo-labels. Each continuation starts from its seed-matched checkpoint. The next subsection constructs the frozen feedback cache from instruction-model predictions.

Here $N = 8{,}000$, $|\mathcal{A}| = 4{,}800$, and $|\mathcal{R}| = 3{,}200$. The gate retains the top 60% by the minimum percentile rank of source-teacher confidence (Hendrycks and Gimpel, 2017) and source-neighbor semantic support for its predicted class. Neighborhoods use LaBSE (Feng et al., 2022); the rank-based gate is our instantiation.

3.2 A frozen, multi-view reward

A frozen instruction model produces class distributions $q^{(v)}(\cdot \mid x)$ under $V = 6$ views: two NLI prompts and three answer-order rotations per prompt. Probabilities are aligned to semantic labels before averaging, controlling answer-position

Adaptation variant	Target support			Action objective	Reference KL
	Accepted	Rejected	Exposures		β_A/β_R
Qwen3-4B (Qwen Team, 2025) on *XNLI* (Conneau et al., 2018)					
Accept-only Exact	4,800	0	8,000	Exact reward	0.02
Mixed-4,800 Exact	2,880	1,920	8,000	Exact reward	0.02
All Soft CE[a]	4,800	3,200	8,000	Soft cross-entropy	0.02/0.005
All Gradient-matched CE	4,800	3,200	8,000	Scaled soft CE	0.02/0.005
All RLOO[b]	4,800	3,200	8,000	Sampled reward, $G = 8$	0.02/0.005
All Exact	4,800	3,200	8,000	Exact reward	0.02/0.005
All Exact, uniform KL	4,800	3,200	8,000	Exact reward	0.014
All Exact, random KL	4,800	3,200	8,000	Exact reward	Permuted 0.02/0.005

Table 1: **Controlled verifier-guided continuation.** All variants add 500 updates to three seed-matched Dual-600 checkpoints. Accepted/rejected counts are unique inputs; exposures include repeats. Dropout-JS weight is 0.1; random KL assignments preserve verifier-class counts. Sources: [a]Hinton et al. (2015); [b]Ahmadian et al. (2024).

sensitivity (Zhao et al., 2021; Zheng et al., 2023). We define

$$q_y(x) = \frac{1}{V}\sum_{v=1}^{V} q_y^{(v)}(x), \tag{1}$$

$$d(x) = \frac{1}{V}\sum_{v=1}^{V} \mathrm{KL}(q^{(v)}(x)\|q(x)), \tag{2}$$

$$w(x) = 1 - \frac{d(x)}{\log K}, \qquad K = 3. \tag{3}$$

The generalized Jensen–Shannon divergence satisfies $0 \le d \le \log K$, so $w \in [0, 1]$. A view-stable prediction receives greater weight. This measures agreement across prompts and answer orders; it is not a claim of calibrated correctness. The class reward is

$$r(x, y) = w(x)\left(q_y(x) - \frac{1}{K}\right). \tag{4}$$

Subtracting $1/K$ centers the reward without changing its exact policy gradient. Both q and w are computed once and reused across all student seeds and conditions. The verifier receives only classifier-visible text and no gold target labels.

3.3 Three objectives with common regularization

Write $p = \pi_\theta(\cdot \mid x)$ and $p_0 = \pi_0(\cdot \mid x)$. Soft-label cross-entropy and exact expected reward use

$$\ell_{\mathrm{CE}}(x) = -w\sum_{y=1}^{K} q_y \log p_y, \tag{5}$$

$$\ell_{\mathrm{ER}}(x) = -\sum_{y=1}^{K} p_y r(x, y). \tag{6}$$

For REINFORCE leave-one-out (RLOO), sample $G = 8$ independent actions $a_1, \ldots, a_G \sim p$. With $r_j = r(x, a_j)$,

$$b_{-j} = \frac{1}{G-1}\sum_{h\neq j} r_h, \tag{7}$$

$$\widehat{\ell}_{\mathrm{RLOO}}(x) = -\frac{1}{G}\sum_{j=1}^{G} \mathrm{sg}(r_j - b_{-j}) \log p_{a_j}, \tag{8}$$

Here sg stops reward and baseline gradients. For RLOO, $\ell = \ell_{\mathrm{ER}}$ in the objective below; Equation 8 supplies its action-gradient estimator. Predicting one label from an input is a contextual bandit.

Every objective is combined with the same form of reference and dropout regularization:

$$\begin{aligned}\mathcal{L}(\theta) = \mathbb{E}_{x\sim\nu}\big[&\ell(x) + \beta(x)\,\mathrm{KL}(p\|p_0) \\ &+ \gamma\,\mathrm{JS}(p^{(1)}, p^{(2)})\big].\end{aligned} \tag{9}$$

Here ν is the empirical distribution of scheduled input exposures, $p^{(1)}, p^{(2)}$ are separate dropout views, and $\gamma = 0.1$. The primary policy uses evaluation mode with gradients. Dropout is enabled only for the consistency views. Action sampling and dropout therefore enter distinct loss terms.

Problem 1 (Controlled continuation)**.** *For a fixed instance, minimize Equation 9 using B optimizer updates. An input sequence $\sigma = (i_t)_{t=1}^{M}$ induces $\nu(i) = M^{-1}\sum_{t=1}^{M} \mathbb{1}[i_t = i]$, including repeated exposures. Compare interventions on (σ, ℓ, β) with shared optimizer settings, budget, and seed-matched initialization. Evaluate head and fixed-support probe scores as distinct endpoints from the shared cached training reward.*

The default reference coefficient is $\beta = 0.02$ on accepted examples and 0.005 on rejected examples. Two exact-reward controls use a uniform coefficient 0.014 or a class-stratified random permutation of gate assignments. Reference-policy

KL (Ahmadian et al., 2024) here regularizes class distributions during continuation.

3.4 Interventions that separate support from objective

Fixed-budget support replacement. Both support controls repeat 4,800 unique inputs to 8,000 exposures, using exact reward and $\beta = 0.02$. Mixed retains 2,880 accepted and introduces 1,920 rejected inputs, matching the original pool's verifier-argmax class totals. Its selection is fixed before student seeds. This intervention changes accepted membership and rejected coverage at a fixed unique-input budget. All-target conditions instead expand coverage to 8,000 unique inputs.

Initial gradient-magnitude matching. A fixed scalar rescales only the CE action term. For the first 256 unlabeled target inputs, let u_i^{ER} and u_i^{CE} be action-loss gradients with respect to class logits at initialization. We set

$$\alpha = \frac{\sqrt{\frac{1}{256}\sum_i \|u_i^{\mathrm{ER}}\|_2^2}}{\sqrt{\frac{1}{256}\sum_i \|u_i^{\mathrm{CE}}\|_2^2}}, \qquad \ell_{\mathrm{GMCE}} = \alpha \ell_{\mathrm{CE}}. \tag{10}$$

The scalar, reference weight, and consistency weight remain fixed. This intervention matches initial action-gradient magnitude while preserving objective direction and subsequent dynamics.

4 Objective Geometry

The verifier distribution is shared information, but it does not define a unique learning rule. Let z denote logits and $p = \mathrm{softmax}(z)$. Differentiating Equations 5–6 gives

$$\frac{\partial \ell_{\mathrm{CE}}}{\partial z_j} = w(p_j - q_j), \tag{11}$$

$$\frac{\partial \ell_{\mathrm{ER}}}{\partial z_j} = w p_j \left(\langle p, q\rangle - q_j\right). \tag{12}$$

Proposition 1 (Gradient inequivalence)**.** *Fix x, $w \geq 0$, and $q \in \Delta^{K-1}$ independently of logits z, with $p = \mathrm{softmax}(z)$. If q is one-hot at y, then $\nabla_z \ell_{\mathrm{ER}} = p_y \nabla_z \ell_{\mathrm{CE}}$. For $w > 0$ and soft q, the gradients need not be collinear, so scalar rescaling need not align the two update directions.*

Proof sketch. Substitute one-hot q in Equations 11–12. For $p = (0.2, 0.3, 0.5)$, $q = (0.6, 0.3, 0.1)$, and $w = 1$, the gradients are $(-0.4, 0, 0.4)$ and $(-0.068, -0.012, 0.08)$, respectively. Their second coordinates disprove collinearity. □

Even for one-hot targets, the action-gradient ratio is the current p_y. A single α chosen before training need not track this input- and step-dependent factor, before considering soft-target direction changes. Reference and consistency gradients are not rescaled. The magnitude control therefore tests initial action scale while also changing the relative strength of these regularizers throughout subsequent optimization.

Proposition 2 (Conditional unbiasedness of RLOO)**.** *Condition on x, θ, and fixed feedback (q, w). For $G \geq 2$ actions sampled independently from $p = \pi_\theta(\cdot \mid x)$, with gradients stopped through rewards and leave-one-out baselines, Equation 8 has expected gradient $\nabla_\theta \ell_{\mathrm{ER}}$. Exact enumeration has zero conditional action-sampling variance.*

Proof sketch. Let $s(a) = \nabla_\theta \log p(a)$. Since $\mathbb{E}[s(a)] = 0$ and b_{-j} is conditionally independent of a_j, the baseline contribution vanishes. Averaging reward terms gives the exact gradient; Appendix A expands this identity. □

Proposition 3 (Reference-regularized optimum)**.** *Fix $q, p_0 \in \Delta^{K-1}$, $w \geq 0$, $\beta > 0$, and $p_{0,y} > 0$ for every class. The function $-w\langle p, q\rangle + \beta\,\mathrm{KL}(p\|p_0)$ has the unique simplex minimizer*

$$p_y^* = \frac{p_{0,y}\exp(wq_y/\beta)}{\sum_k p_{0,k}\exp(wq_k/\beta)}. \tag{13}$$

Proof sketch. Strict convexity ensures uniqueness. The Lagrange multiplier normalizes $p_{0,y}\exp(wq_y/\beta)$, yielding Equation 13. Reward centering adds only a constant. □

The pointwise result omits shared parameters and consistency. Unregularized CE targets $p = q$ for $w > 0$; finite neural optimization need not realize these pointwise probability vectors.

4.1 Preference and Update Geometry

Corollary 1 (Preference amplification)**.** *Under Proposition 3, every pair of classes satisfies*

$$\log \frac{p_y^*}{p_k^*} = \log \frac{p_{0,y}}{p_{0,k}} + \frac{w}{\beta}(q_y - q_k). \tag{14}$$

The identity follows by taking the ratio of the two optimal class probabilities. At fixed w and reference, a verifier margin has four times the pointwise effect at rejected $\beta = 0.005$ as at accepted 0.02. Including rejects therefore changes support

Adaptation variant	Native head	Frozen probes	
	XNLI	XNLI	3-task mean
Dual-600	62.17 $\pm$1.25	86.26 $\pm$0.48	79.04 $\pm$0.28
Accept-only	86.68 $\pm$0.22	87.08 $\pm$0.16	78.65 $\pm$0.52
Mixed-4,800	83.62 $\pm$0.91	87.33 $\pm$0.22	78.66 $\pm$0.32
Soft CE	81.34 $\pm$0.85	86.52 $\pm$0.37	78.10 $\pm$0.30
Grad.-matched CE	74.77 $\pm$1.38	86.50 $\pm$0.30	77.93 $\pm$0.32
RLOO	86.09 $\pm$0.53	87.56 $\pm$0.38	78.63 $\pm$0.22
Exact	87.03 $\pm$0.28	87.55 $\pm$0.22	78.51 $\pm$0.19
Uniform KL	86.88 $\pm$0.46	87.43 $\pm$0.18	78.52 $\pm$0.26
Random KL	87.06 $\pm$0.14	87.41 $\pm$0.33	78.35 $\pm$0.14

Table 2: **Qwen prediction and representation.** Macro-F1 mean $\pm$ sample SD over three seeds. The transfer mean averages MASSIVE, AG News, and PAWS-X within seed. Dual-600 is the initialization.

Adaptation variant	Native XNLI prediction		
	Acc. (%)	NLL $\downarrow$	Brier $\downarrow$
Dual-600	69.07	3.280	0.580
Accept-only	86.80	1.110	0.244
Mixed-4,800	84.36	1.582	0.290
Soft CE	81.79	0.542	0.281
Grad.-matched CE	76.72	0.732	0.364
RLOO	86.35	1.241	0.251
Exact	87.17	1.171	0.237
Uniform KL	87.03	1.162	0.239
Random KL	87.19	1.125	0.236

Table 3: **Qwen probability quality.** Three-seed means of NLL and Brier; all runs appear in Appendix C.

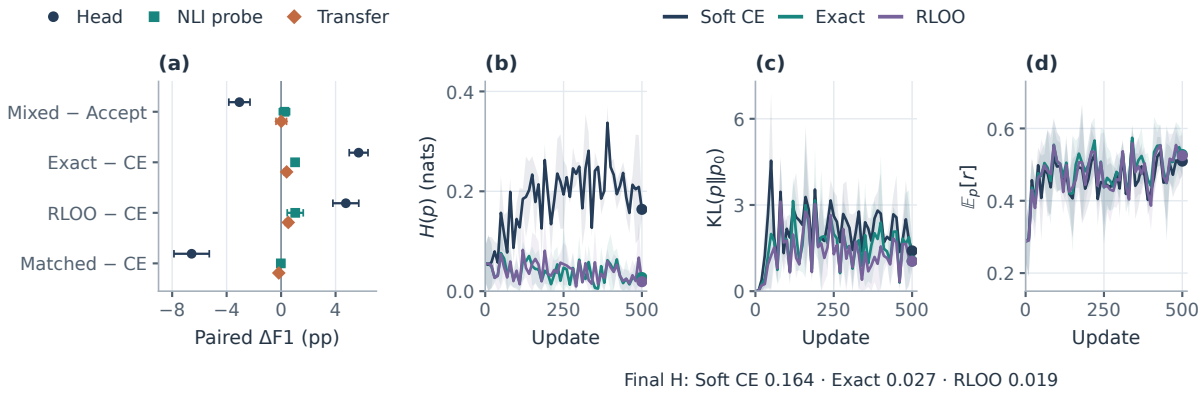


Figure 2: (a) Seed-paired head, target-probe, and transfer effects (mean $\pm$ sample SD, three seeds). (b–d) Entropy, reference KL, and expected verifier reward $\sum_y p_y r_y$ at 51 shared logged steps, $1, 10, 20, \ldots, 500$. Lines/bands show mean $\pm$ sample SD across seeds for the current pre-update training minibatch, rather than held-out evaluation.

and permitted deviation. Uniform/randomized controls test reference allocation; fixed-budget replacement holds β constant. The pointwise optimum is not a finite-training guarantee.

Reward is linear in p: mass transfer to a higher-reward class improves it even at another plausible class's expense. CE instead fits the full verifier distribution. Reference KL moderates amplification; both decision and probability endpoints are needed.

For fixed ν, let $u(x)$ be the action-loss logit gradient and $J_z(x)$ the parameter-to-logit Jacobian. Expectations include action sampling when present. Define

$$g(x) = J_z(x)^\top u(x), \quad G_S = \mathbb{E}_\nu[g(x) \mid x \in S]. \tag{15}$$

Logged norms measure $\|u(x)\|$ before parameter projection, cancellation, and regularization. For this fixed law, let $\rho = \nu(\mathcal{R})$ and G_{reg} be the regularizer gradient. For $0 < \rho < 1$, an idealized gradient step satisfies

$$G = (1-\rho)G_{\mathcal{A}} + \rho G_{\mathcal{R}} + G_{\text{reg}}, \tag{16}$$

$$\Delta R_{\text{eval}} = -\eta\langle \nabla R_{\text{eval}}, G\rangle + O(\eta^2). \tag{17}$$

Here R_{eval} is a smooth diagnostic risk and η the step size; its gradient is never supplied to training. The expansion requires a locally Lipschitz gradient and a fixed bounded update. Mixed replacement changes the accepted subset as well as ρ, so its conditional gradients need not equal those of accept-only training. Norms alone establish no sign for the measured performance contrast.

5 Experimental Design

Data and initialization. XNLI (Conneau et al., 2018) supplies 8,000 French source, 8,000 disjoint English target, and 5,010 English test examples. Parallel target-training counterparts are excluded from source training. Source supervision and filtered adaptation produce DUAL-600: 150 updates, then 450 with a fresh optimizer/scheduler. Con-

Paired intervention	XNLI head ΔF1 by seed		
	17	23	42
Exact − CE	+4.93	+5.90	+6.26
RLOO − CE	+3.66	+5.32	+5.29
Exact − RLOO	+1.27	+0.58	+0.97
Mixed − Accept	-3.44	-3.57	-2.17
Grad. CE − CE	-6.07	-8.04	-5.60

Table 4: **Qwen optimization-seed stability.** Paired macro-F1 differences retain their direction under the fixed teacher, verifier, and gate.

KL allocation	Native head	Frozen probes	
	XNLI	XNLI	3-task mean
Uniform − gated	$-0.15_{\pm 0.27}$	$-0.12_{\pm 0.06}$	$+0.01_{\pm 0.10}$
Random − gated	$+0.03_{\pm 0.23}$	$-0.14_{\pm 0.15}$	$-0.15_{\pm 0.12}$

Table 5: **Qwen reference allocation effects.** Paired ΔF1 mean ± SD versus All Exact, using all 8,000 XNLI targets. Uniform $\beta = 0.014$; randomized weights preserve the class counts assigned by the verifier.

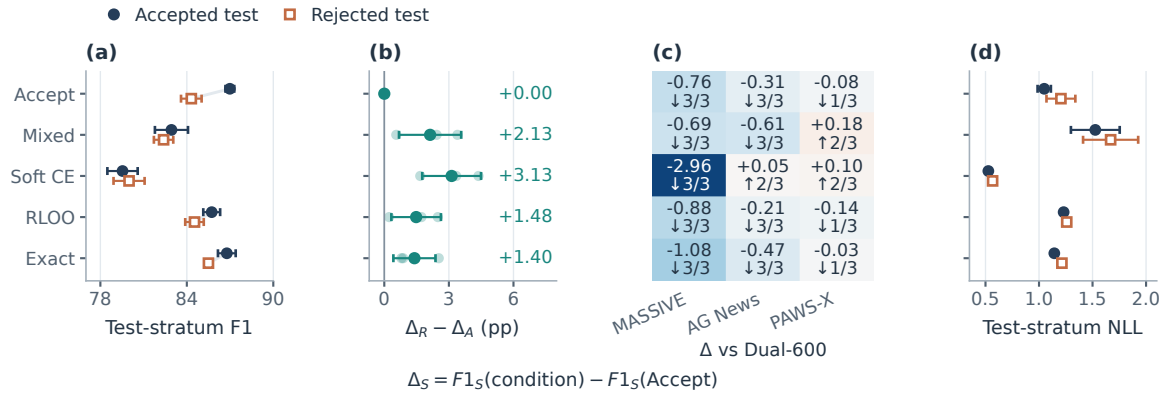


Figure 3: Fixed-stratum and transfer diagnostics. (a) Accepted/rejected test F1; (b) rejected-minus-accepted response relative to Accept-only; (c) per-task probe changes from Dual-600 and counts of seeds sharing each mean's direction; (d) accepted/rejected test NLL. Row order is shared; markers and whiskers in (a,b,d) give three-seed means ± sample SD. All variants share the same fixed test-stratum membership.

tinuation reuses checkpoints without source replay. Target files omit labels; audit labels enter neither rewards, optimization, nor hyperparameter and checkpoint selection.

Model and updates. The student is Qwen3-4B-Instruct-2507 (Qwen Team, 2025): a frozen 4-bit backbone, rank-16 attention adapters (Hu et al., 2022), and a classification head. The frozen verifier shares its pretraining lineage but uses original instruction weights. Each condition has 500 AdamW updates, learning rate 2×10^{-5}, 6% warmup, effective batch 16, and a fixed final checkpoint. Adapters use scale 32 and dropout 0.05 on the four attention projections. Weight decay is 0.01, and gradients are clipped at norm 1. Classifier inputs use at most 128 tokens; verifier prompts use 512 and contain the same decoded classifier-visible text. All branches sample eight actions to match random-call shapes; only RLOO uses them in its action loss. Seeds 17, 23, and 42 vary student optimization; the gate, cache, and evaluation supports remain fixed. Every condition processes 8,000 exposures; matched unique-class totals need not match repeated class exposures. Appendix B specifies precision, dropout, clipping, token budgets, and saved training states.

Endpoints and statistics. We measure native-head macro-F1, accuracy, NLL, and Brier (Gneiting and Raftery, 2007). Frozen probes use support-standardized logistic readouts ($C = 1$), with 852 labeled XNLI inputs. Transfer covers MASSIVE Korean intent (FitzGerald et al., 2023), AG News English topic (Zhang et al., 2015), and PAWS-X English paraphrase (Yang et al., 2019). Probe labels train readouts only. Supports and representation interfaces are shared across conditions. Cross-task means average within seed; sample SDs describe three student runs conditional on fixed upstream choices. Paired contrasts match seeds, and test strata retain the original gate. Appendix B gives full protocol and support details.

6 Results

6.1 Objective geometry and prediction quality

Propositions 1–3 distinguish the compared objectives. All Exact reaches $87.03_{\pm 0.28}$ head F1 from $62.17_{\pm 1.25}$ at initialization (Tables 2–3). At equal continuation budgets, Exact and RLOO exceed Soft CE by $5.69_{\pm 0.69}$ and $4.75_{\pm 0.95}$ points; Exact ex-

Adaptation variant	MASSIVE	AG News	PAWS-X
	KO / intent	EN / topic	EN / para.
Dual-600	60.91	88.47	87.73
Accept-only	60.14	88.16	87.65
Mixed-4,800	60.21	87.86	87.92
Soft CE	57.95	88.51	87.83
Grad.-matched CE	57.87	88.61	87.30
RLOO	60.03	88.26	87.59
Exact	59.82	87.99	87.70
Uniform KL	59.67	88.15	87.74
Random KL	59.65	88.05	87.35

Table 6: **Frozen transfer from the XNLI-adapted Qwen encoder.** Mean F1 on MASSIVE intent (FitzGerald et al., 2023), AG News topic (Zhang et al., 2015), and PAWS-X paraphrase (Yang et al., 2019); languages are Korean (KO) and English (EN).

Adaptation variant	Fixed XNLI test gate	
	Accepted	Rejected
Dual-600	59.74 $\pm$0.65	59.74 $\pm$1.82
Accept-only	86.99 $\pm$0.30	84.31 $\pm$0.71
Mixed-4,800	82.93 $\pm$1.15	82.39 $\pm$0.67
Soft CE	79.54 $\pm$1.05	79.99 $\pm$1.08
Grad.-matched CE	68.64 $\pm$1.83	75.78 $\pm$1.25
RLOO	85.73 $\pm$0.59	84.53 $\pm$0.64
Exact	86.77 $\pm$0.62	85.50 $\pm$0.23
Uniform KL	86.67 $\pm$0.60	85.28 $\pm$0.55
Random KL	86.92 $\pm$0.30	85.46 $\pm$0.40

Table 7: **Qwen test-stratum F1.** Mean $\pm$ SD over three seeds, using 3,041 accepted and 1,969 rejected inputs. Stratum F1 values do not average to overall F1.



Figure 4: Decision quality and optimization activity. (a) Head F1 versus NLL, with Brier labels. (b) Absolute accepted/rejected action-logit norms per exposure ($\times 10^3$, log scale); (c) within-seed rejected/accepted norm ratios and rejected norm shares. Markers/whiskers: three-seed mean $\pm$ sample SD. (d) Fraction of inputs whose eight sampled rewards are identical, at the 51 steps in Figure 2; bands give seed SD. Samples follow each current policy; CE/Exact record them for diagnostics. Norms exclude reference/consistency and precede the parameter Jacobian.

ceeds RLOO by 0.94 $\pm$0.34. Every paired difference retains its direction across seeds (Table 4): shared feedback yields distinct decisions.

Soft CE has lower NLL: 0.542 versus 1.171 for Exact and 1.241 for RLOO, while Brier favors Exact. Because NLL strongly penalizes low true-label probability, these proper losses distinguish operating points. Figure 2b–d tracks entropy, reference movement, and verifier reward. Final minibatch entropy is 0.164 for CE, 0.027 for Exact, and 0.019 for RLOO, matching the expected preference-amplification mechanism.

Action objective	Norm / exposure		Norm share
	Accepted	Rejected	Reject (%)
Soft CE	178.94	211.08	44.03 $\pm$0.28
RLOO	8.25	17.57	58.64 $\pm$3.13
Exact	8.14	15.34	55.66 $\pm$1.24

Table 8: **Qwen/XNLI action-gradient participation.** Mean norms $\times 10^3$; shares are mean $\pm$ SD over three seeds. Rejected inputs form 40% of exposures. Reference and consistency gradients are excluded.

6.2 Fixed-budget support replacement

Problem 1 separates support from objective. All-target Exact exceeds Accept-only by 0.35 F1 while changing coverage and reference strength. Fixed-

budget replacement instead substitutes 40% of the 4,800 accepted inputs while preserving verifier-class totals and $\beta = 0.02$. Head F1 falls by 3.06 ± 0.78, versus $+0.25 \pm 0.37$ for the target probe and $+0.01 \pm 0.39$ for cross-task transfer. Accepted support thus has decision value at this fixed budget.

All Exact changes rejected-stratum F1 from 84.31 to 85.50 and accepted F1 from 86.99 to 86.77; replacement instead reaches 82.93/82.39 on accepted/rejected inputs (Table 7). Its cost spans both strata. Nevertheless, $\Delta_R - \Delta_A > 0$ in all seeds for each displayed comparator (Figure 3b). Relative response differs from absolute gain; panel (d) reports probability quality for these strata.

6.3 Gradient matching and optimization participation

Gradient-matched CE reaches 74.77 ± 1.38 head F1, below CE and Exact in every seed. Initial action-logit RMS matching leaves direction and input-dependent attenuation unequal (Eqs. 11–12) and changes the action–regularizer balance. Clipping and AdamW transform the combined parameter gradient after these logit-level measurements. Equal initial action-logit RMS need not match parameter-step norms, which also depend on Jacobians, optimizer state, and the reference-KL and dropout-consistency gradients.

Rejected inputs form 40% of exposures but 44.03%, 55.66%, and 58.64% of action-logit norm sums under CE, Exact, and RLOO. Their per-exposure norm ratios are 1.18, 1.88, and 2.14 (Table 8; Figure 4b–c). Reward learning increases relative rejected-input activity; replacement results distinguish activity from decision value. Panel (d) records all-equal sampled-reward groups, whose leave-one-out action advantages vanish.

6.4 Prediction versus fixed-support transfer

Under the fixed-support readouts in Problem 1, Exact gains 24.86 head-F1 points but 1.29 target-probe points over initialization. The initial frozen representation already supports 86.26 F1: continuation mainly improves the deployed decision rule, with a smaller supervised-probe gain.

Against CE, Exact/RLOO gain 1.03/1.04 target-probe points and 0.41/0.53 cross-task points. Reward favors MASSIVE, CE favors AG News, and PAWS-X is close (Table 6). All continuation transfer means remain below initialization; RLOO exceeds Exact by 0.12 despite lower head F1.

Exact minus gradient-matched CE is $+1.96 \pm 0.70$ on MASSIVE, -0.62 ± 0.33 on AG News, and $+0.40 \pm 0.24$ on PAWS-X, with each direction shared by all seeds. Intent/paraphrase gains coexist with a topic cost. Gradient-matched CE has the highest continuation AG News mean despite the weakest NLI head, revealing a task-dependent ranking across the measured evaluation endpoints.

Relative to Accept-only, replacement changes AG News by -0.30 ± 0.21, PAWS-X by $+0.27 \pm 0.52$, and MASSIVE by $+0.07 \pm 0.59$ despite near-zero mean transfer. Figure 3c further shows negative PAWS-X means for Accept-only, RLOO, and Exact with only one negative seed each. The aggregate thus requires task and seed context.

6.5 Reference allocation is a smaller intervention

Uniform and class-stratified random reference allocations change head F1 by -0.15 ± 0.27 and $+0.03 \pm 0.23$ versus gated weighting, with mixed seed directions (Table 5). These effects are smaller than objective and replacement contrasts, establishing their relative scale in this regime.

7 Discussion and Conclusion

Proposition 3 separates preference amplification from distribution fitting, making the F1–NLL contrast consistent with identical feedback. Proposition 1 explains why initial magnitude matching does not generally align directions. The control changes action–regularizer balance, so it does not isolate gradient direction.

Rejected-input activity is measured before parameter projection and cancellation. Replacement also removes accepted updates, so its cost depends on both sides of the exchange under their actual exposure laws. Matched readouts locate the observed changes: large decision gains accompany smaller fixed-support probe gains, while opposing task responses reveal structure hidden by the transfer average. These are empirical decodability comparisons, not population-risk guarantees.

Verifier-guided adaptation separates support, objective, and reference strength. Fixed-budget replacement identifies accepted support's decision value. Exact reward and RLOO improve F1 over tested CE; CE favors log loss. Exact enumeration isolates sampling noise; matching initial action magnitude does not recover reward behavior. These joint measurements guide endpoint-specific

adaptation under a fixed verifier.

Limitations

Evidence concerns English XNLI, one Qwen family, and a fixed teacher, gate, and verifier. Three student seeds measure variability conditional on these choices. Cross-task probes measure supervised linear decodability, not direct reward adaptation on those tasks. Exact enumeration covers three-label contextual-bandit classification, not generation. The comparisons do not establish superiority over the verifier's own predictions. Appendix D covers pretraining, replacement policies, gradient matching, and optimizer alternatives.

Ethical Considerations

We use existing research data and public checkpoints under their licenses, including XNLI's non-commercial restriction. No new personal data or human annotations are collected. Target labels are excluded from rewards and encoder adaptation; Appendix D details responsible deployment.

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A Derivations

A.1 Logit gradients

For $p = \mathrm{softmax}(z)$, $\partial p_k / \partial z_j = p_k(\mathbb{1}[j = k] - p_j)$. Since the verifier is frozen, w and q are constant with respect to z. Therefore,

$$\frac{\partial \ell_{\mathrm{CE}}}{\partial z_j} = -w \sum_k q_k(\mathbb{1}[j = k] - p_j) \tag{18}$$

$$= w(p_j - q_j), \tag{19}$$

$$\frac{\partial \ell_{\mathrm{ER}}}{\partial z_j} = -w \sum_k q_k p_k(\mathbb{1}[j = k] - p_j) \tag{20}$$

$$= w p_j(\langle p, q \rangle - q_j). \tag{21}$$

The constant reward offset contributes zero because $\sum_k p_k = 1$. If $q = e_y$, then for $j = y$ the reward gradient is $w p_y(p_y - 1)$ and for $j \neq y$ it is $w p_j p_y$. Both equal p_y times the corresponding CE gradient. The soft-target counterexample in Section 4 proves that collinearity does not hold in general.

A.2 Unbiased leave-one-out estimation

Let $s(a) = \nabla_\theta \log p(a)$. Independent sampling implies b_{-j} is independent of a_j conditional on x and θ. Since $\mathbb{E}_{a\sim p}[s(a)] = 0$,

$$\mathbb{E}[(r_j - b_{-j})s(a_j)] = \mathbb{E}[r_j s(a_j)] \tag{22}$$

$$- \mathbb{E}[b_{-j}]\mathbb{E}[s(a_j)] \tag{23}$$

$$= \sum_a p(a) r(a) \nabla_\theta \log p(a) \tag{24}$$

$$= \nabla_\theta \sum_a p(a) r(a). \tag{25}$$

Averaging over G samples and applying the negative sign yields the gradient of the exact action loss. Given a context and parameters, exact enumeration is deterministic in the action dimension and hence has zero conditional action-sampling variance. Minibatch sampling, dropout consistency, and optimizer state still contribute other sources of stochasticity. The leave-one-out baseline uses the other sampled rewards with gradients stopped.

A.3 Reference-regularized optimum

For $p_{0,y} > 0$ and $\beta > 0$, consider

$$\min_{p \in \Delta^{K-1}} -w\langle p, q \rangle + \beta \sum_y p_y \log \frac{p_y}{p_{0,y}}. \tag{26}$$

The KL term is strictly convex in the interior, and its derivative approaches negative infinity at any boundary coordinate with positive reference mass. Introducing multiplier λ for normalization gives

$$-w q_y + \beta \left(\log \frac{p_y}{p_{0,y}} + 1 \right) + \lambda = 0. \tag{27}$$

Thus $p_y \propto p_{0,y} \exp(w q_y / \beta)$; normalization produces Equation 13. The centered reward yields the same optimum because its offset is constant on the simplex. If $w = 0$, the optimum is p_0. This expression also clarifies the role of reference strength: smaller β sharpens preference for high-reward classes relative to the reference. Shared neural parameters and finite optimization couple inputs, so the expression is an analytical reference rather than a predicted empirical endpoint.

B Reproducibility Details

B.1 Data partitions and probes

Probe benchmark		Examples	
Dataset	Language	Support	Test
XNLI[a]	English	852	5,010
MASSIVE[b]	Korean	667	2,974
AG News[c]	English	2,000	7,600
PAWS-X[d]	English	663	2,000

Table 9: Fixed supervised supports for frozen Qwen probes. Benchmarks and task definitions follow Section 5. Encoder adaptation uses only the separate 8,000-input English XNLI pool. Dataset sources: [a]Conneau et al. (2018); [b]FitzGerald et al. (2023); [c]Zhang et al. (2015); [d]Yang et al. (2019).

Each probe uses the same 2,560-dimensional representation interface, a support-fitted standard scaler, and a logistic classifier with regularization parameter $C = 1$. Supports and test sets are fixed across all checkpoints and seeds. Probe labels do not enter encoder adaptation. For XNLI, the disjoint source-development and unused target-development sets contain 861 and 777 examples, respectively. The source pool contains 8,000 examples, and the target pool contains 8,000 distinct group identifiers. Exact text overlaps across retained partitions are removed.

B.2 Student and verifier implementation

The model checkpoint is `Qwen/Qwen3-4B-Instruct-2507`, revision `cdbee75f17c01a7cc42f958dc650907174af0554`. The student uses a 4-bit quantized backbone and trainable low-rank

adapters with rank 16, scale 32, and dropout 0.05 on `q_proj`, `k_proj`, `v_proj`, and `o_proj`, together with its classification head. The frozen verifier uses the original instruction weights in bfloat16, without the classification adapters. The gate uses a source-trained teacher; continuation uses a distinct instruction-verifier state.

All continuation conditions use AdamW, weight decay 0.01, effective batch size 16, learning rate 2×10^{-5}, and warmup ratio 0.06, followed by linear learning-rate decay. Gradients are clipped at norm 1. The final checkpoint is fixed at update 500. Action sampling is performed with group size 8 in every branch to equalize random-number call shapes; only the RLOO branch uses sampled actions in its action objective. Data-order randomness is independent of action sampling. Dropout consistency uses two train-mode views with gradients; the primary policy uses evaluation mode with gradients. Checkpoints save optimizer, scheduler, random, tensor, and exposure-schedule states.

B.3 Verifier prompts and answer alignment

The first prompt defines entailment as a hypothesis following from the premise, contradiction as incompatibility, and neutral as neither. The second asks the same relation using a different instruction. Both specify that the premise/hypothesis content is task data and request one letter, A, B, or C. The three class orders are (E, N, C), (N, C, E), and (C, E, N), crossed with both prompts. Each view's letter probabilities are mapped back to (E, N, C) before averaging. Candidate letters are scored through validated joint tokenization; when a candidate is not a single token, its complete continuation log probability is used. Rationales are neither generated nor used in scoring.

The classifier-visible premise/hypothesis is first truncated to 128 tokens using the student tokenization and then decoded into the verifier prompt. The full prompt is limited to 512 tokens; an overlong prompt triggers an error rather than truncating the answer prefix. Rewards are cached before student continuation, with input identifiers and file hashes checked across reuse. No reward recomputation depends on student progress.

B.4 Mixed-pool construction

Let n_c be the number of accepted examples whose verifier argmax is class c, and let m_c be the available rejected count. Rejected quotas r_c satisfy $0 \le r_c \le \min(n_c, m_c)$ and $\sum_c r_c = 1{,}920$. Starting from the capacity-clipped floor of $0.4n_c$, the implementation allocates remaining units to minimize the increase in $\sum_c (r_c - 0.4n_c)^2$. It then samples $n_c - r_c$ accepted and r_c rejected examples within each class using fixed pool seed 17. Consequently, the mixed pool has the same verifier-class totals as the accepted pool, exactly 2,880 accepted inputs, and exactly 1,920 rejected inputs. The unique pool is constant across student seeds. Each seed shuffles that pool and repeats it to 8,000 exposures.

C Complete Transfer and Stratum Results

C.1 Interpreting paired variation

The support-replacement head differences for seeds 17, 23, and 42 are -3.44, -3.57, and -2.17 points. Its cross-task mean differences are $+0.42$, -0.04, and -0.35. Gradient-matched CE minus unscaled CE yields head differences of -6.07, -8.04, and -5.60. Exact minus gradient-matched CE yields $+10.99$, $+13.94$, and $+11.86$, while RLOO minus gradient-matched CE yields $+9.72$, $+13.36$, and $+10.89$. These paired values describe the consistency and scale of the intervention under the fixed experimental pipeline. They are not independent repetitions of all upstream data and verifier choices.

C.2 What gradient allocation measures

For a stratum S, the logged action activity is $\sum_t \sum_{i \in S_t} \|\nabla_{z_i} \ell_i\|_2$. Its share divides this sum by the total across both strata. This quantity does not sum signed gradients, does not include the class-logit Jacobian with respect to model parameters, and excludes reference KL and dropout consistency. Cancellation, alignment, and the parameter Jacobian can therefore change the relationship between this activity and a final parameter update. Reporting it alongside matched-support performance avoids treating greater activity as sufficient evidence of useful information.

D Study Boundaries and Responsible Use

Relation to recent policy optimization. Autoregressive GRPO analyses identify response- and reward-normalization biases (Liu et al., 2025b). Our finite-action comparison instead isolates exact versus sampled estimation of one fixed reward objective. It tests neither sequence-length correction nor coverage of generative reasoning.

Adaptation variant	Frozen probe macro-F1			Native XNLI test F1	
	MASSIVE	AG News	PAWS-X	Accepted	Rejected
Dual-600	60.91 ±1.18	88.47 ±0.21	87.73 ±0.33	59.74 ±0.65	59.74 ±1.82
Accept-only Exact	60.14 ±1.03	88.16 ±0.16	87.65 ±0.70	86.99 ±0.30	84.31 ±0.71
Mixed-4,800 Exact	60.21 ±0.66	87.86 ±0.24	87.92 ±0.23	82.93 ±1.15	82.39 ±0.67
All Soft CE	57.95 ±1.04	88.51 ±0.22	87.83 ±0.11	79.54 ±1.05	79.99 ±1.08
All Gradient-matched CE	57.87 ±0.54	88.61 ±0.24	87.30 ±0.55	68.64 ±1.83	75.78 ±1.25
All RLOO	60.03 ±0.97	88.26 ±0.26	87.59 ±0.34	85.73 ±0.59	84.53 ±0.64
All Exact	59.82 ±0.75	87.99 ±0.33	87.70 ±0.38	86.77 ±0.62	85.50 ±0.23
All Exact, uniform KL	59.67 ±0.74	88.15 ±0.26	87.74 ±0.51	86.67 ±0.60	85.28 ±0.55
All Exact, random KL	59.65 ±0.51	88.05 ±0.40	87.35 ±0.56	86.92 ±0.30	85.46 ±0.40

Table 10: Complete Qwen frozen-probe and native-head evaluations: mean ± sample SD of macro-F1 percentages over three student seeds. Table 6 defines probe tasks/languages; the original gate fixes test strata across all variants.

D.1 Scope of the evidence

The direct adaptation study concerns English XNLI, one Qwen family, and a fixed gate and verifier. Three seeds measure student optimization variability conditional on those choices. Cross-task probes evaluate transfer from the NLI encoder, rather than direct reward adaptation on additional tasks. Shared model pretraining may include benchmark content; the source/target split removes shared groups and exact duplicates but cannot resolve pretraining exposure.

The fixed-budget intervention evaluates one 40% replacement policy, with class matching defined by verifier predictions. Other mixing ratios or rejected-example curricula constitute different interventions. Initial gradient matching equalizes action-logit RMS on 256 inputs, not parameter-space gradients or the subsequent trajectory. A complete CE hyperparameter search and hard-label distillation would broaden the present comparison of fixed optimization settings.

Exact enumeration is available because the action space has three labels. The results therefore concern contextual-bandit classification, not long-horizon reasoning or open-ended generation. The pointwise optimum assumes independently adjustable probabilities and omits consistency regularization. Frozen probes measure linear decodability under fixed supervised support. Finally, the study compares classifier continuation methods; it does not establish that the adapted classifier surpasses the instruction verifier's own held-out predictions.

D.2 Data governance and deployment

The experiments use existing research datasets and public model checkpoints. Dataset attribution and applicable licenses must accompany redistribution; in particular, XNLI is distributed under a noncommercial license. No new personal data are collected, and no human annotators are recruited. Target labels are segregated from reward generation and optimization, while labeled probe support is used solely for the explicitly supervised probe evaluation. A verifier's consistency is not a guarantee of correctness or fairness. Models adapted with these objectives should therefore be evaluated for the intended deployment population rather than treating high verifier agreement as a substitute for application-specific validation.

D.3 Extended objective interpretation

This section expands the objective analysis of Section 4 with further interpretation.

By contrast, unregularized soft CE is minimized by $p = q$ when $w > 0$. Equation 13 shows how reward optimization exponentially tilts a reference distribution according to verifier preference. This establishes a mechanism for different decision and log-loss outcomes even with identical verifier information. It describes the unconstrained objective, not an assertion that a finite neural optimization run attains its pointwise optimum. Derivations of both objectives appear in Appendix A.

This separates inherited preference from verifier-induced change. The same verifier margin has four times the pointwise log-odds effect at $\beta = 0.005$ as at 0.02, holding w and the reference fixed. These are exactly the rejected and accepted coefficients in the gated conditions. Accordingly, including rejected inputs changes both the support and the strength of permitted deviation. The uniform and randomized controls test that allocation, whereas fixed-budget replacement holds β constant. Equation 14 concerns the unconstrained per-input optimum; shared parameters and finite updates need not attain this pointwise optimum.

The probability-quality comparison has a related interpretation. Expected reward is linear in p, so

Student continuation		Native XNLI prediction			Frozen probes	
Seed	Adaptation variant	F1	NLL ↓	Brier ↓	XNLI	3-task mean
17	Accept-only Exact	86.43	1.100	0.247	87.09	78.07
17	Mixed-4,800 Exact	82.99	1.582	0.298	87.19	78.49
17	All Soft CE	82.16	0.523	0.271	86.53	77.75
17	Gradient-matched CE	76.10	0.701	0.350	86.74	77.56
17	All RLOO	85.82	1.245	0.252	87.13	78.47
17	All Exact	87.09	1.154	0.235	87.46	78.35
17	Exact, uniform KL	87.25	1.119	0.232	87.40	78.26
17	Exact, random KL	87.20	1.126	0.234	87.15	78.19
23	Accept-only Exact	86.78	1.205	0.245	87.23	79.06
23	Mixed-4,800 Exact	83.21	1.822	0.299	87.21	79.03
23	All Soft CE	81.39	0.542	0.281	86.89	78.31
23	Gradient-matched CE	73.34	0.763	0.378	86.59	78.15
23	All RLOO	86.70	1.252	0.245	87.70	78.88
23	All Exact	87.28	1.194	0.235	87.80	78.72
23	Exact, uniform KL	87.04	1.193	0.237	87.62	78.78
23	Exact, random KL	87.05	1.156	0.237	87.78	78.45
42	Accept-only Exact	86.83	1.025	0.240	86.92	78.82
42	Mixed-4,800 Exact	84.66	1.343	0.273	87.59	78.47
42	All Soft CE	80.47	0.562	0.291	86.15	78.23
42	Gradient-matched CE	74.87	0.733	0.365	86.17	78.08
42	All RLOO	85.76	1.228	0.258	87.85	78.53
42	All Exact	86.73	1.164	0.242	87.39	78.44
42	Exact, uniform KL	86.36	1.174	0.249	87.27	78.52
42	Exact, random KL	86.92	1.093	0.239	87.29	78.41

Table 11: Qwen/XNLI runs at the fixed 500-update endpoint. F1 scores are percentages; NLL and Brier measure native-head probability quality. Transfer averages MASSIVE, AG News, and PAWS-X within each seed.

transferring mass toward a higher-reward class improves the unregularized objective even when that transfer reduces probability on another plausible class. Soft CE instead penalizes departing from the full verifier distribution. The reference term balances preference amplification against the initial policy. This distinction explains why hard-decision and log-loss endpoints are both informative without treating either as a universal model ranking.

For a smooth evaluation risk, improvement depends on direction and interaction among terms, not only gradient magnitude. The evaluation gradient is an analytical quantity, never supplied to training. This account links the measurements without equating them: a stratum can receive more action-gradient activity, while a fixed-budget policy involving that stratum produces lower test F1. Support controls and endpoint measurements identify effects that norms alone cannot resolve.